In [7]:
```
!pip install transformers seqeval -q

import numpy as np
import torch
import nltk
from nltk.corpus import treebank
from datasets import Dataset, DatasetDict
from transformers import (
    AutoTokenizer,
    AutoModelForTokenClassification,
    TrainingArguments,
    Trainer,
    DataCollatorForTokenClassification,
)
from seqeval.metrics import accuracy_score, f1_score, classification_report
import random

nltk.download('treebank', quiet=True)


MODEL_NAME = "prajjwal1/bert-mini"
MAX_LEN = 128
BATCH_SIZE = 32
EPOCHS = 5
LEARNING_RATE = 3e-4
WEIGHT_DECAY = 0.01
OUTPUT_DIR = "/kaggle/working/bert_pos_output"

sentences = list(treebank.tagged_sents())
random.shuffle(sentences)

all_tags = sorted(set(tag for sent in sentences for _, tag in sent))
label2id = {tag: i for i, tag in enumerate(all_tags)}
id2label = {i: tag for tag, i in label2id.items()}
num_labels = len(all_tags)

print(f"Total sentences: {len(sentences)}")
print(f"Number of POS labels: {num_labels}")

def sentences_to_dict(sents):
    return {
        "tokens": [[w for w, _ in s] for s in sents],
        "pos_tags": [[label2id[t] for _, t in s] for s in sents],
    }

train_sents = sentences[:int(0.8 * len(sentences))]
val_sents   = sentences[int(0.8 * len(sentences)):int(0.9 * len(sentences))]
test_sents  = sentences[int(0.9 * len(sentences)):]

dataset = DatasetDict({
    "train":      Dataset.from_dict(sentences_to_dict(train_sents)),
    "validation": Dataset.from_dict(sentences_to_dict(val_sents)),
    "test":       Dataset.from_dict(sentences_to_dict(test_sents)),
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})

print(dataset)


tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)

def tokenise_and_align(examples):
    tokenised = tokenizer(
        examples["tokens"],
        truncation=True,
        max_length=MAX_LEN,
        is_split_into_words=True,
    )
    all_labels = []
    for i, pos_tags in enumerate(examples["pos_tags"]):
        word_ids = tokenised.word_ids(batch_index=i)
        labels = []
        prev_word_id = None
        for word_id in word_ids:
            if word_id is None:
                labels.append(-100)
            elif word_id != prev_word_id:
                labels.append(pos_tags[word_id])
            else:
                labels.append(-100)
            prev_word_id = word_id
        all_labels.append(labels)
    tokenised["labels"] = all_labels
    return tokenised

tokenised_dataset = dataset.map(
    tokenise_and_align,
    batched=True,
    remove_columns=["tokens", "pos_tags"],
)


model = AutoModelForTokenClassification.from_pretrained(
    MODEL_NAME,
    num_labels=num_labels,
    id2label=id2label,
    label2id=label2id,
)


def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)

    true_labels = [
        [id2label[l] for l in label_seq if l != -100]
        for label_seq in labels
```

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    ]
    pred_labels = [
        [id2label[p] for p, l in zip(pred_seq, label_seq) if l != -100]
        for pred_seq, label_seq in zip(predictions, labels)
    ]

    return {
        "accuracy": accuracy_score(true_labels, pred_labels),
        "f1_macro": f1_score(true_labels, pred_labels, average="macro"),
    }


training_args = TrainingArguments(
    output_dir=OUTPUT_DIR,
    num_train_epochs=EPOCHS,
    per_device_train_batch_size=BATCH_SIZE,
    per_device_eval_batch_size=BATCH_SIZE,
    learning_rate=LEARNING_RATE,
    weight_decay=WEIGHT_DECAY,
    eval_strategy="epoch",
    save_strategy="epoch",
    load_best_model_at_end=True,
    metric_for_best_model="f1_macro",
    logging_steps=50,
    warmup_ratio=0.1,
    fp16=torch.cuda.is_available(),
    report_to="none",
)

data_collator = DataCollatorForTokenClassification(tokenizer)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenised_dataset["train"],
    eval_dataset=tokenised_dataset["validation"],
    processing_class=tokenizer,
    data_collator=data_collator,
    compute_metrics=compute_metrics,
)

trainer.train()

predictions, labels, _ = trainer.predict(tokenised_dataset["test"])
predictions = np.argmax(predictions, axis=-1)

true_labels = [
    [id2label[l] for l in label_seq if l != -100]
    for label_seq in labels
]
pred_labels = [
    [id2label[p] for p, l in zip(pred_seq, label_seq) if l != -100]
    for pred_seq, label_seq in zip(predictions, labels)
```

```
]

print("\n--- seqeval Report ---")
print(classification_report(true_labels, pred_labels))
print(f"Token-level Accuracy : {accuracy_score(true_labels, pred_labels):.4f}"
print(f"Macro F1             : {f1_score(true_labels, pred_labels, average='ma


def predict(sentence: str):
    words = sentence.split()
    inputs = tokenizer(
        words,
        is_split_into_words=True,
        return_tensors="pt",
        truncation=True,
        max_length=MAX_LEN,
    )
    word_ids = tokenizer(
        words,
        is_split_into_words=True,
        truncation=True,
        max_length=MAX_LEN,
    ).word_ids()
    inputs = {k: v.to(model.device) for k, v in inputs.items()}
    with torch.no_grad():
        logits = model(**inputs).logits
    preds = torch.argmax(logits, dim=-1)[0].tolist()
    seen = set()
    result = []
    for idx, word_id in enumerate(word_ids):
        if word_id is None or word_id in seen:
            continue
        seen.add(word_id)
        result.append((words[word_id], id2label[preds[idx]]))
    return result

print("\nSample prediction:")
print(predict("The quick brown fox is running quickly"))
```

```
Total sentences: 3914
Number of POS labels: 46
DatasetDict({
    train: Dataset({
        features: ['tokens', 'pos_tags'],
        num_rows: 3131
    })
    validation: Dataset({
        features: ['tokens', 'pos_tags'],
        num_rows: 391
    })
    test: Dataset({
        features: ['tokens', 'pos_tags'],
        num_rows: 392
    })
})
Map:   0%|          | 0/3131 [00:00<?, ? examples/s]
Map:   0%|          | 0/391 [00:00<?, ? examples/s]
Map:   0%|          | 0/392 [00:00<?, ? examples/s]
Loading weights:   0%|          | 0/69 [00:00<?, ?it/s]
BertForTokenClassification LOAD REPORT from: prajjwal1/bert-mini
Key                                         | Status     |
--------------------------------------------+------------+-
cls.predictions.transform.dense.bias        | UNEXPECTED |
cls.predictions.bias                        | UNEXPECTED |
cls.seq_relationship.weight                 | UNEXPECTED |
bert.pooler.dense.bias                      | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |
bert.embeddings.position_ids                | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight  | UNEXPECTED |
bert.pooler.dense.weight                    | UNEXPECTED |
cls.predictions.decoder.bias                | UNEXPECTED |
cls.predictions.transform.dense.weight      | UNEXPECTED |
cls.predictions.decoder.weight              | UNEXPECTED |
cls.seq_relationship.bias                   | UNEXPECTED |
classifier.weight                           | MISSING    |
classifier.bias                             | MISSING    |

Notes:
- UNEXPECTED        :can be ignored when loading from different task/architectu
re; not ok if you expect identical arch.
- MISSING        :those params were newly initialized because missing from the
checkpoint. Consider training on your downstream task.
warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps` inst
ead.
/usr/local/lib/python3.12/dist-packages/torch/autograd/function.py:583: UserWar
ning: Was asked to gather along dimension 0, but all input tensors were scalar
s; will instead unsqueeze and return a vector.
  return super().apply(*args, **kwargs)  # type: ignore[misc]
```

[245/245 00:14, Epoch 5/5]

Epoch	Training Loss	Validation Loss	Accuracy	F1 Macro
1	No log	1.123600	0.903623	0.599171
2	4.078524	0.505021	0.938302	0.701843
3	0.754278	0.414128	0.947930	0.766135
4	0.435864	0.382541	0.951760	0.772804
5	0.337357	0.369358	0.952588	0.776811

```
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: NNS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
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1: UserWarning: RB seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
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/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: -NONE- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: . seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: `` seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: IN seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: DT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: PRP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: , seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: EX seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: MD seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: VB seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: NN seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: TO seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: PRP$ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: CC seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
```

```
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: : seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBG seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: JJ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: '' seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: NNP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBZ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: WRB seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: CD seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: WP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: POS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: NNPS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: WDT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: $ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: JJS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: JJR seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: RP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: UH seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
```

```
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: RBR seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: -LRB- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: -RRB- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: LS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: RBS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: PDT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

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/usr/local/lib/python3.12/dist-packages/torch/autograd/function.py:583: UserWar
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1: UserWarning: WP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: POS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: NNPS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
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/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: RP seems not to be NE tag.
```

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/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: NNP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: VBZ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: WRB seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: CD seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: WP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: POS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: NNPS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: WDT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: $ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: JJS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: JJR seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: RP seems not to be NE tag.
```

```
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: UH seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: RBR seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: -LRB- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: -RRB- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: LS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: RBS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: PDT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
There were missing keys in the checkpoint model loaded: ['bert.embeddings.Layer
Norm.weight', 'bert.embeddings.LayerNorm.bias', 'bert.encoder.layer.0.attentio
n.output.LayerNorm.weight', 'bert.encoder.layer.0.attention.output.LayerNorm.bi
as', 'bert.encoder.layer.0.output.LayerNorm.weight', 'bert.encoder.layer.0.outp
ut.LayerNorm.bias', 'bert.encoder.layer.1.attention.output.LayerNorm.weight',
'bert.encoder.layer.1.attention.output.LayerNorm.bias', 'bert.encoder.layer.1.o
utput.LayerNorm.weight', 'bert.encoder.layer.1.output.LayerNorm.bias', 'bert.en
coder.layer.2.attention.output.LayerNorm.weight', 'bert.encoder.layer.2.attenti
on.output.LayerNorm.bias', 'bert.encoder.layer.2.output.LayerNorm.weight', 'ber
t.encoder.layer.2.output.LayerNorm.bias', 'bert.encoder.layer.3.attention.outpu
t.LayerNorm.weight', 'bert.encoder.layer.3.attention.output.LayerNorm.bias', 'b
ert.encoder.layer.3.output.LayerNorm.weight', 'bert.encoder.layer.3.output.Laye
rNorm.bias'].
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.La
yerNorm.beta', 'bert.embeddings.LayerNorm.gamma', 'bert.encoder.layer.0.attenti
on.output.LayerNorm.beta', 'bert.encoder.layer.0.attention.output.LayerNorm.gam
ma', 'bert.encoder.layer.0.output.LayerNorm.beta', 'bert.encoder.layer.0.outpu
t.LayerNorm.gamma', 'bert.encoder.layer.1.attention.output.LayerNorm.beta', 'be
rt.encoder.layer.1.attention.output.LayerNorm.gamma', 'bert.encoder.layer.1.out
put.LayerNorm.beta', 'bert.encoder.layer.1.output.LayerNorm.gamma', 'bert.encod
er.layer.2.attention.output.LayerNorm.beta', 'bert.encoder.layer.2.attention.ou
tput.LayerNorm.gamma', 'bert.encoder.layer.2.output.LayerNorm.beta', 'bert.enco
der.layer.2.output.LayerNorm.gamma', 'bert.encoder.layer.3.attention.output.Lay
erNorm.beta', 'bert.encoder.layer.3.attention.output.LayerNorm.gamma', 'bert.en
coder.layer.3.output.LayerNorm.beta', 'bert.encoder.layer.3.output.LayerNorm.ga
mma'].
/usr/local/lib/python3.12/dist-packages/torch/autograd/function.py:583: UserWar
ning: Was asked to gather along dimension 0, but all input tensors were scalar
s; will instead unsqueeze and return a vector.
  return super().apply(*args, **kwargs)  # type: ignore[misc]
```

```
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: DT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: NN seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: IN seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: JJ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBZ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBN seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: -NONE- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: NNS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: WDT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: CC seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: . seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: , seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: WRB seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: NNP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: CD seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: POS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: PRP$ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
```

```
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBD seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: TO seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: $ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VB seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: VBG seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: RB seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: JJR seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: PRP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: : seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: RP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: PDT seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: `` seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: '' seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: MD seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: NNPS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: -LRB- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: -RRB- seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:171: UserWarning: JJS seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
```

```
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: WP seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: RBR seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: WP$ seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: EX seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/sequence_labeling.py:17
1: UserWarning: # seems not to be NE tag.
  warnings.warn('{} seems not to be NE tag.'.format(chunk))
/usr/local/lib/python3.12/dist-packages/seqeval/metrics/v1.py:57: UndefinedMetr
icWarning: Precision and F-score are ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this behavi
or.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
--- seqeval Report ---
              precision    recall  f1-score   support

           '       1.00      1.00      1.00        67
           B       0.93      0.91      0.92       497
          BD       0.97      0.99      0.98       300
          BG       0.90      0.88      0.89       140
          BN       0.95      0.96      0.95       218
          BP       0.93      0.96      0.95       135
          BR       0.00      0.00      0.00        10
          BZ       1.00      0.97      0.99       207
           C       0.97      0.93      0.95       209
           D       0.99      0.97      0.98       366
          DT       1.00      0.91      0.95        53
           J       0.91      0.86      0.88       552
          JR       0.81      0.95      0.87        40
          JS       1.00      0.95      0.97        19
           N       0.93      0.94      0.93      1691
          NP       0.86      0.89      0.87       553
         NPS       0.00      0.00      0.00        26
          NS       0.98      0.95      0.97       582
           O       1.00      1.00      1.00       212
          OS       0.99      1.00      0.99        83
           P       0.83      0.82      0.82        49
          P$       0.00      0.00      0.00         3
          RB       1.00      0.77      0.87        26
          RP       0.86      0.97      0.91       161
         RP$       1.00      0.94      0.97        77
           T       0.96      0.97      0.96       825
           X       1.00      0.83      0.91         6
           _       1.00      1.00      1.00      1097
           `       1.00      1.00      1.00        67

   micro avg       0.95      0.94      0.94      8271
   macro avg       0.85      0.84      0.84      8271
weighted avg       0.94      0.94      0.94      8271

Token-level Accuracy : 0.9585
Macro F1             : 0.8443

Sample prediction:
[('The', 'DT'), ('quick', 'JJ'), ('brown', 'NN'), ('fox', 'NN'), ('is', 'VBZ'),
('running', 'VBG'), ('quickly', 'RB')]
```

In [ ]: